



Liver & Extrahepatic Biliary Apparatus

Histology GIT Block · MEDucation by Mattin Majid

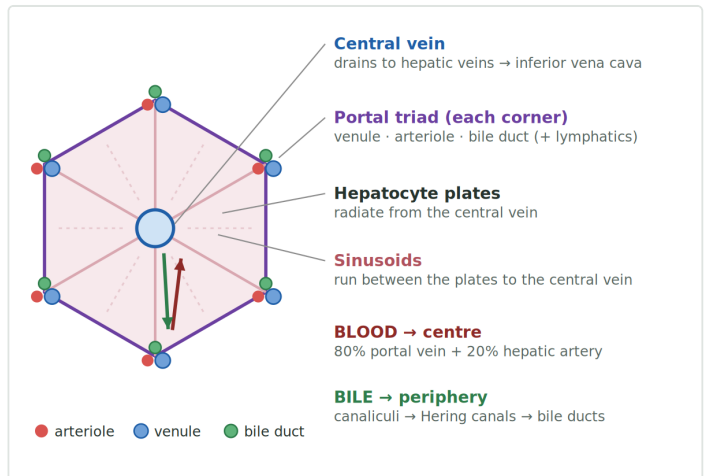
BEFORE YOU READ – 3 BLIND ID CHALLENGES

1. Draw a classic lobule: what sits in the centre and what sits at the corners?
2. Which liver cell lives in the space of Disse and stores vitamin A — and which one phagocytoses old red cells?
3. Trace bile from a hepatocyte to the duodenum, naming each duct.

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 · The liver at low power — one-line ID rules

- Liver = **largest gland of the body**, under the diaphragm; two major lobes + two smaller inferior lobes. Covered by a thin **capsule (Glisson's)** and the mesothelium of the visceral peritoneum.
- The capsule thickens at the **hilum**: **portal vein and hepatic artery enter; right and left hepatic ducts and lymphatics leave**.
- **Parenchyma** = hexagonal **lobules** of hepatocytes radiating from a **central vein**, with **portal triads** at the corners.



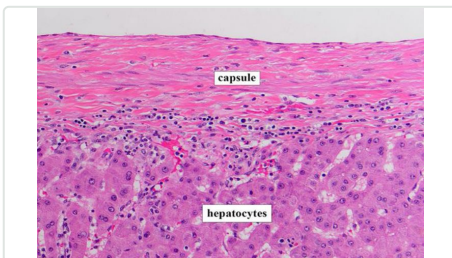
Anchor diagram: one classic lobule — blood flows in to the centre, bile flows out to the corners.

STRUCTURE	ID ON THE SLIDE	WHAT IT IS
Lobule	hexagonal ; hepatocytes radiate outward from a central vein	the liver parenchyma
Central vein	thin-walled vessel in the centre of the lobule	receives the sinusoid blood
Portal triad	at each corner (3–6 per lobule in humans): a bile duct , an arteriole (hepatic artery branch), a venule (portal vein branch), lymphatic vessels — all in connective tissue	the whole CT-wrapped group is the portal canal

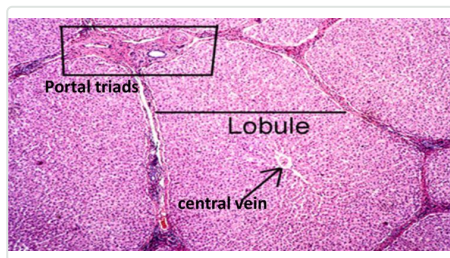
Slide wording: "triad" names three structures but the slide lists four (the lymphatic vessels are the 4th), and it gives both "six corners" and "3 to 6 portal triads per lobule". Learn it as: hexagonal lobule, central vein in the middle, a portal triad at the corners.

MAGNIFICATION LADDER (SCAN-PATH ORGANISATION OF THE LECTURE'S FEATURES – THE LECTURE GIVES NO MAGNIFICATIONS)

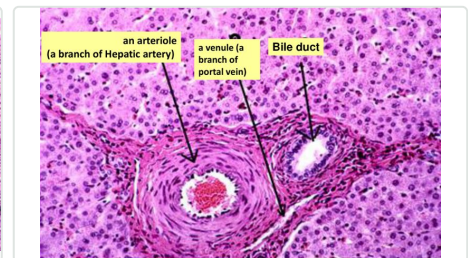
MAG	WHAT TO LOOK FOR
×4	Capsule on the surface; hexagonal lobules with a central vein in the middle and portal triads at the corners.
×10	Portal canal: bile duct (cuboidal to columnar), arteriole, venule in CT; plates of hepatocytes radiating from the central vein with sinusoids between them.
×40	Hepatocytes (large, polyhedral, eosinophilic cytoplasm); Kupffer cells on the sinusoid lumen; the narrow space of Disse behind the endothelium.



Capsule and underlying hepatocytes.



Lobule: portal triads at the corners, central vein in the middle.



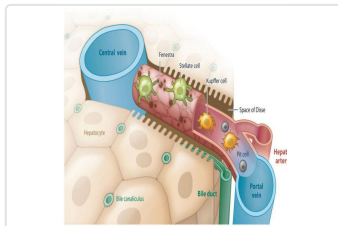
Portal triad: arteriole (hepatic artery branch), venule (portal vein branch), bile duct.

2 · Cells and hepatocyte surfaces

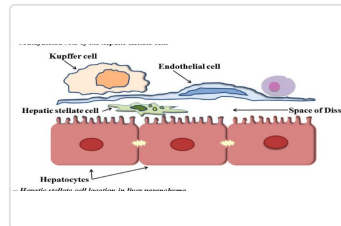
CELL	WHERE	ID / KEY FEATURE	JOB
Hepatocyte	plates that anastomose into a network; sinusoid on two sides of every cell	large polyhedral cell with 6+ surfaces; eosinophilic cytoplasm rich in mitochondria and organelles	primary structural and functional unit; secretes bile
Sinusoid endothelium	lines the sinusoids between the plates	irregular dilated vessels; discontinuous, fenestrated endothelium	blood passes to the hepatocytes
Kupffer cell	luminal surface of the sinusoidal endothelium	monocyte-derived ; ovoid nucleus; usually holds engulfed material	phagocytose aged red cells (heme, iron → reuse or ferritin storage); antigen-presenting , clear bacteria and debris from portal blood
Ito (stellate) cell	space of Disse	lipid droplets in the cytoplasm; long contractile processes	stores vitamin A ; major source of collagen / matrix in liver injury ; regulates sinusoidal flow; cytokines regulate Kupffer cells

Look-alike pair: Kupffer vs Ito. **Kupffer = on the lumen side of the endothelium, phagocytic; Ito = in the space of Disse, lipid-droplet (vitamin A) store that lays down collagen in injury.** The **space of Disse** is the narrow perisinusoidal space between the hepatocyte and the endothelium; hepatocyte **microvilli** project into it.

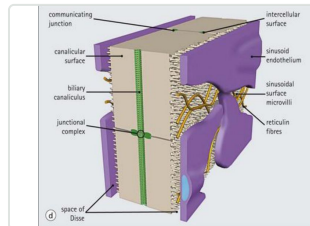
HEPATOCTE SURFACE	FACES	FEATURE / JOB
Sinusoidal	the blood sinusoid	microvilli project into the space of Disse → exchange with the blood
Canalicular	a neighbouring hepatocyte, forming a groove	longitudinal depressions of adjacent cells form the wall of a bile canaliculus ; irregular microvilli; primary site of bile secretion
Intercellular	adjacent hepatocytes	area of contact with cell junctions ; exchange between cells



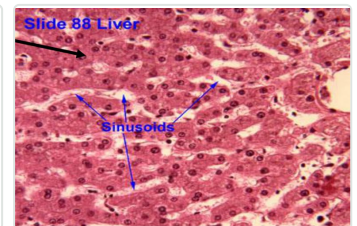
Sinusoid: central vein, fenestrae, stellate cell, Kupffer cell, space of Disse, bile canaliculus, portal vein, hepatic artery, bile duct.



Kupffer cell, endothelial cell, hepatic stellate cell, hepatocytes.



Hepatocyte surfaces: canalicular, sinusoidal, intercellular; space of Disse.

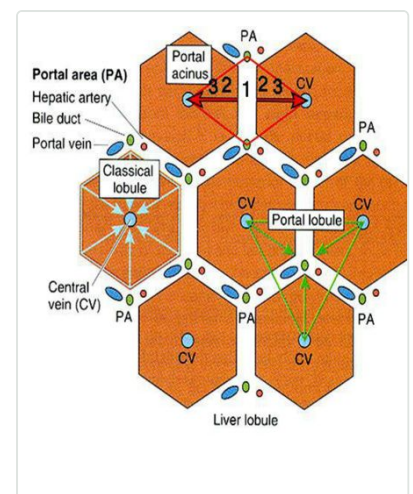


Slide: sinusoids between hepatocyte plates.

3 · Three ways to divide the liver — the discriminator grid

	CLASSIC LOBULE	PORTAL LOBULE	HEPATIC ACINUS
Based on	direction of blood flow	direction of bile flow (opposite to blood)	oxygen, nutrient, toxin changes along the sinusoids (functional)
Shape	hexagon	triangle	diamond
Centre / corners	central vein in the centre; portal triad at each corner	portal triad at the centre ; a central vein at each of 3 corners	2 central veins + 2 portal triads define the 4 corners

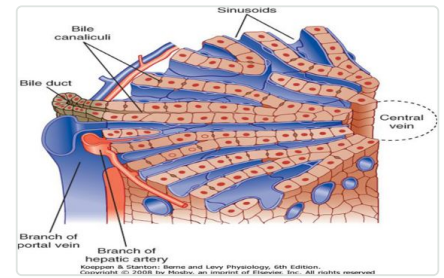
- **Acinus zones:** a line joining the two portal triads splits the diamond in two triangles, each divided into **zones I-III** by distance from the terminal distributing vessels.
- **Zone I** (nearest them) = **highest oxygen, nutrients and toxins; Zone III** = nearest the central vein.



Classic lobule, portal lobule, portal acinus (zones 1-3).

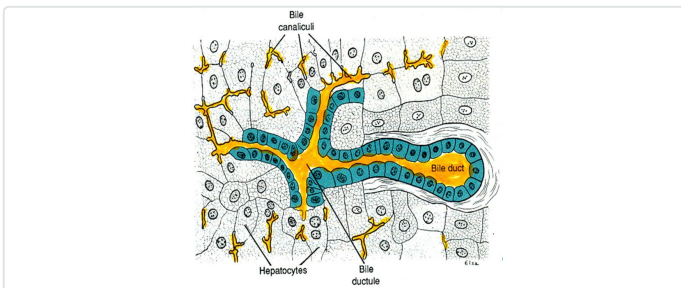
4 • Blood in, bile out

- **Blood: 80% portal vein** (from pancreas, spleen, intestine; oxygen-poor, nutrient-rich) + **20% hepatic artery** (higher oxygen) → portal venules / hepatic arterioles in the triad → **sinusoids** → **central vein** → **hepatic veins (right + left)** → **inferior vena cava**.
- **Bile** = exocrine secretion of hepatocytes: water, electrolytes, **bile acids**, phospholipids, cholesterol, **bilirubin**. About **90%** derived from intestinal absorption; ~**10%** of bile acid synthesized in the hepatocyte **smooth ER**. Bile acids **emulsify lipids** for lipase digestion and absorption.

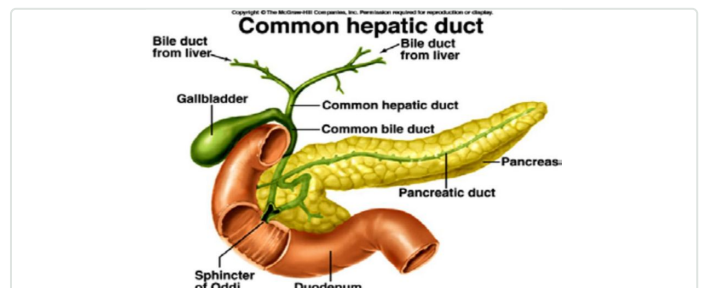


Sinusoids, central vein, bile canaliculi, branches of portal vein and hepatic artery.

STEP	DUCT	LINING / NOTE
1	Bile canaliculi	limited only by the plasma membrane of two hepatocytes
2	Bile ductules (canals of Hering)	cuboidal
3	Bile ducts in the portal triad	cuboidal to columnar
4	Right and left hepatic ducts → common hepatic duct (hilum)	join the cystic duct from the gallbladder → common bile duct
5	Common bile duct + main pancreatic duct	form the common hepatopancreatic duct = ampulla of Vater , opens into the duodenum, regulated by the sphincter of Oddi



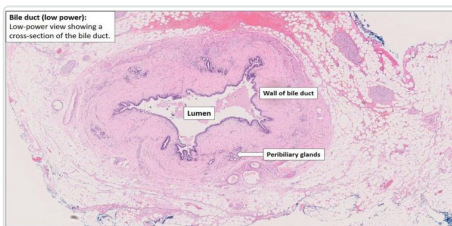
Bile canaliculus → ductule → bile duct; hepatocytes.



Extrahepatic ducts, gallbladder, pancreas, sphincter of Oddi.

5 • Extrahepatic ducts and the gallbladder

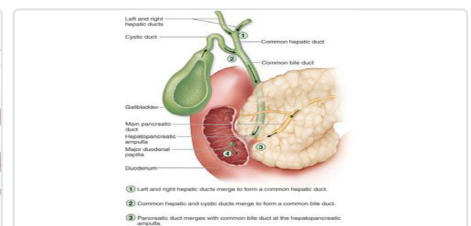
	EXTRAHEPATIC BILE DUCTS	GALLBLADDER
Lining	simple columnar mucous membrane (hepatic, cystic and common bile ducts)	highly folded mucosa, simple columnar with microvilli , no goblet cells ; lamina propria beneath
Muscle	thin muscularis , thicker near the duodenum and forming the sphincter of Oddi in the duodenal papilla (regulates bile flow)	two or three irregular smooth-muscle layers
Other layers	—	submucosa like the GI tract (vessels, lymphatics, nerves); serosa (mesothelium), or adventitia where it touches the liver
Job	carry bile to the duodenum	pear-shaped sac under the liver; stores and concentrates bile , releases it via the bile duct; bile regulated by cholecystokinin (CCK)



Bile duct, low power: lumen and wall.



Gallbladder: highly folded mucosa.



Ducts joining: hepatic + cystic → common bile duct; ampulla; duodenal papilla.

Pearls — what must survive if everything else is forgotten

- **Lobule:** central vein in the middle, **portal triads** (bile duct + arteriole + venule, plus lymphatics) at the corners; hepatocyte plates radiate, with a **sinusoid on two sides** of every cell.
- **Blood** = 80% portal vein + 20% hepatic artery → sinusoids → central vein → hepatic veins → IVC. **Bile flows the opposite way:** canaliculi → canals of Hering → bile ducts.
- **Kupffer** = on the sinusoid lumen, phagocytoses aged red cells. **Ito** = space of Disse, vitamin A droplets, collagen in injury.
- **Three models:** classic lobule (blood flow, hexagon) · portal lobule (bile flow, triangle) · acinus (oxygen gradient, diamond; **Zone I** nearest the triads = richest in oxygen, nutrients, toxins).
- **Gallbladder:** folded simple columnar mucosa, microvilli, **no goblet cells**; stores and concentrates bile. Bile ducts: simple columnar; muscle thickens into the **sphincter of Oddi** (ampulla of Vater).

Test yourself — transfer-task MCQs

Unseen fields, not slide recall. Commit to an answer before you uncover the key.

1. At the corner of a liver lobule a pathologist finds a small duct lined by cuboidal to columnar cells beside an arteriole and a venule in connective tissue. What is this structure, and which of its vessels carries about 80% of the liver's blood?

- A. Central vein; the hepatic vein
- B. Portal triad; the portal vein branch
- C. Portal triad; the hepatic artery branch
- D. Sinusoid; the central vein
- E. Space of Disse; the hepatic artery branch

2. A macrophage-like cell on the luminal side of the sinusoidal endothelium contains engulfed red-cell debris. Which cell is it, and what becomes of the iron it frees?

- A. Ito cell; excreted in bile
- B. Kupffer cell; reused or stored in ferritin
- C. Hepatocyte; converted to bile acids
- D. Kupffer cell; stored in lipid droplets as vitamin A
- E. Endothelial cell; secreted into the space of Disse

3. In chronic liver injury a cell in the space of Disse lays down large amounts of collagen. What is this cell, and what does it normally store?

- A. Kupffer cell; iron
- B. Hepatocyte; glycogen
- C. Ito (stellate) cell; vitamin A
- D. Endothelial cell; cholesterol
- E. Ito (stellate) cell; ferritin

4. In a hepatic acinus, hepatocytes nearest the portal triads are compared with those nearest the central vein. Which statement is correct?

- A. Zone I sinusoid blood has the highest oxygen, nutrient and toxin concentration
- B. Zone III sinusoid blood has the highest oxygen concentration
- C. All zones receive blood of identical composition
- D. Zone I lies nearest the central vein
- E. Zone II lies nearest the portal triads

5. A biopsy of an extrahepatic bile duct is examined near the duodenum. Which description fits?

- A. Stratified squamous epithelium with a thick muscularis throughout
- B. Simple columnar epithelium; the thin muscularis thickens toward the duodenum into the sphincter of Oddi
- C. Highly folded mucosa without goblet cells and three irregular muscle layers
- D. Cuboidal epithelium limited by two hepatocyte membranes
- E. Fenestrated endothelium over a space of Disse

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1. **B** — Duct + arteriole + venule in CT = portal triad; the portal vein supplies ~80% of hepatic blood.
- 2. **B** — Kupffer cells phagocytose aged red cells; the iron is reused or stored in ferritin.
- 3. **C** — Ito cells store vitamin A in lipid droplets and make collagen in liver injury.
- 4. **A** — Zone I is nearest the triads: highest oxygen, nutrients and toxins; Zone III is nearest the central vein.
- 5. **B** — Bile ducts: simple columnar epithelium; the thin muscularis thickens near the duodenum into the sphincter of Oddi.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Histology GIT Block · MEDucation by Mattin Majid · Source: GIT Histology SGL 4, Dr Hero K. Mustafa
Study alongside the lecture, not instead of it.